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International Perspectives

User Perception and Use of the Academic Library: A Correlation Analysis☆

Jong-Ae Kim

Department of Library and Information Science, Kyonggi University, 154-42 Gwanggyosan-ro Yeongtong-gu, Suwon-si, Gyeonggi-do, Republic of Korea

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ABSTRACT

This study explores user perception, the use of an academic library, and the correlation between user perception and library use. The findings indicate that the academic library is a preferred place to study and spend time during non-class times on campus. The frequent user activities in the library included individual study, reading, computer use, finding information, group study, using reference and information services, and meeting and socializing. The correlation between library use and each of the user activities was examined. The findings suggest that users perceive the academic library primarily as a place for learning and information seeking. The study shows how users perceive and use the academic library, and provides insights into how libraries can strengthen their place and advance their role in the academic community.

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INTRODUCTION

Academic libraries have played a crucial role on campus as the "heart of the university." As an effort to advance their role in the academic community, libraries have accommodated changes in use trends to better support users in the process of learning and intellectual pursuit. Changes in the approaches of users to information seeking and learning have brought the need for transition in library facilities and services. Academic libraries have embraced new technologies and redesigned their services to accommodate changes in user needs and behavior.

Although extensive information is available remotely, the physical library is still a preferred place for users (Applegate, 2009). A range of studies have reported the increase in library facility use (Houlihan, 2005; McKay, 2011; Shill & Tonner, 2004). Researchers have referred to the increase as a "post-internet bounce" (Albanese, 2003). Academic libraries are still one of the most visited places on campus. They support a wide range of user activities for information seeking, individual and group study, socializing, and relaxation. As more social and collaborative aspects of learning have been emphasized in higher education, academic libraries have evolved into centers for social activities as well as centers for intellectual pursuit.

An academic library that adapts to the new academic and social activity trends of users will continue to serve as an essential institution within its community. As such, academic libraries should make continuous efforts to provide user-centered facilities and services based on the understanding of user needs and behavior. Capturing changes in user perception and behavior is critical in planning library facilities and services. Given the importance of academic libraries in academic institutions and the changing nature of user perception and behavior, it is meaningful to examine users' perceptions of academic libraries and how they are related to their use of the libraries. This study explores user perception and the use of an academic library by means of a survey. It investigates user perception, use patterns, and the correlation of user perception and library use. An empirical study on user perception and usage behavior may provide insights into improving management decisions in planning library facilities and services.

LITERATURE REVIEW

Several studies have reported usage statistics and user preferences of academic libraries. In a study comparing pre-project and post-completion facility usage patterns for library facility projects, most of the institutions that completed a library facility project reported overall usage increases (Shill & Tonner, 2004). The specific facility features that were related to increase in usage included the quality of instruction lab, institution type, layout, number of data ports, public access workstations, natural lighting, user work space, telecommunication infrastructure, and overall ambience. The study provided empirical evidence that students would use a comfortable, well-equipped library even though there are electronic resources available remotely. The findings suggest that a well-designed library would attract students as a learning and socializing place and remain essential to the academic community. Martell (2008) reported increase in gate count and the use of electronic networked resources in a study providing data on library use from the

☆ The researcher was a visiting researcher in the College of Information Studies at the University of Maryland when this study was conducted.

E-mail address: jongaekim@kgu.ac.kr.

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Table 1
Preferred places to study and read.

Place	The first preferred place		The second preferred place		The third preferred place	
	Frequency	Percentage	Frequency	Percentage	Frequency	Percentage
Home	65	46.1	22	15.6	29	20.6
Library	61	43.3	55	39.0	17	12.1
Classroom	5	3.5	33	23.4	33	23.4
Lab	4	2.8	9	6.4	8	5.7
Café	4	2.8	15	10.6	29	20.6
Dorm	1	0.7	0	0.0	4	2.8
Lawn and outdoor bench	1	0.7	6	4.3	16	11.3
Other	0	0.0	1	0.7	5	3.5
Total	141	100.0	141	100.0	141	100.0

1990s to 2006. Dotson and Garris (2008) reported an overall increase in library use in a study examining usage patterns in Thomas Cooper Library (TCL) at the University of South Carolina. Specifically, there was increase in the use of group tables, individual tables, and internet-accessible computers and decrease in the use of stacks, connected individual desks, and catalog computers. As the study of building usage indicated a high use of study space, especially of group tables, TCL made more space for group tables and group seating. This resulted in increase in the use of study tables, showing that students wanted study space. The building survey provided the library with an opportunity to examine use of the library facility and user preferences regarding library resources, services, and space.

Applegate (2009) examined student preferences for study space and found that the library is the preferred place for academic activity that is not immediately connected with a class session. The most preferred study space in the university library was a study room, implying that students might prefer to be allowed to talk while studying in the library. The traditional study carrel appeared to have appeal, showing increasing use during the course of a semester. Wireless Internet access and electrical outlets were also considered as important elements for laptop users. Given (2007) examined the perceptions of both faculty and librarians on the design and availability of campus spaces. The study identified the need for welcoming spaces on campus to support students' informational activities and learning processes. The faculty and academic librarians noted that soft chairs, spacious tables, ergonomic workstations, clean surfaces, and aesthetic details were welcoming elements to support academic work. They also emphasized the importance of offering students a social space for collaboration, information sharing, and down-time with peers. Peterson (2005) identified five types of user activity for which library space should be designed: information seeking, recreation, teaching and learning, connection, and contemplation. She explains that library space should accommodate users' activities of seeking a wide range of information for their work and personal lives, and support those who seek entertainment material or activities for recreational purposes. Teaching and learning spaces are also needed to support activities such as group study, class, one-to-one teaching and learning, tutoring, literacy activities, training, and homework help. For connection, libraries serve as communal gathering spaces for those who have academic, social, and cultural pursuits. For those who need a silent space, libraries serve as sanctuaries and designated places for contemplation.

Various studies have examined the relationship between user characteristics and patterns of use of academic libraries. In a study focusing on whether first-year college students' use of academic libraries was associated with the students' socioeconomic characteristics, Soria, Nackerud, and Peterson (2015) found that students' use of academic libraries was associated with a wide variety of socioeconomic indicators. The findings suggested that students from lower socioeconomic status backgrounds were less likely to use the libraries in areas such as circulation, reference services, accessing e-books and e-journals, and using library workstations. Whitmire (2003) examined whether there were differences in undergraduates' academic library use across racial groups. There were similarities across racial groups in terms of the frequency of library activities in which students are engaged, but differences were found in the frequency of activities such as finding materials while browsing the stacks, checking citations, and checking out books. There were also differences in the factors associated with academic library use across culturally diverse racial groups. Hall and Kapa (2015) investigated the specific needs of a library's campus community and found differences in library use across the users from different programs of study. The study reported that only 35% of the respondents came to the library to borrow material, whereas 84% visited the library to use study space. It was also found that the students who used the service points most frequently were not necessarily those who used the study spaces the most. Bridges (2008) examined the differences in undergraduate students' use of the physical library and the virtual library by academic discipline and class standing. The study found significant differences in the use of the physical library and the virtual library by academic discipline. However, no significant differences were determined in the use of the physical library and the virtual library by class standing.

Some studies have examined the relationship between library use and academic performance to demonstrate the positive impact of

Table 2
Preferred places to spend time during non-class times on campus.

Place	The first preferred place		The second preferred place		The third preferred place	
	Frequency	Percentage	Frequency	Percentage	Frequency	Percentage
Library	47	33.3	37	26.2	20	14.2
Café	44	31.2	30	21.3	27	19.1
Classroom	16	11.3	32	22.7	29	20.6
Gym	11	7.8	11	7.8	13	9.2
Lawn and outdoor bench	8	5.7	11	7.8	21	14.9
Lab	7	5.0	12	8.5	19	13.5
Dorm	4	2.8	1	0.7	4	2.8
Other	4	2.8	2	1.4	6	4.3
Sport turf	0	0.0	5	3.5	2	1.4
Total	141	100.0	141	100.0	141	100.0

Table 3
Academic library use (n = 141).

Use of the library		Frequency	Percentage
Frequency of library visits	Almost every day	42	29.8
	A few times a week	50	35.5
	A few times a month	25	17.7
	About once a month	10	7.1
	A few times a year	5	3.5
	About once a year	0	0.0
	Seldom	9	6.4
Average length of each visit to the library	Less than an hour	18	12.8
	One to two hours	48	34.0
	Two to four hours	52	36.9
	Four to six hours	17	12.1
	More than six hours	6	4.3

academic libraries on the academic community. In a study investigating the correlation between students' library usage and their academic performance, Wong and Webb (2011) found a positive correlation between students' book loans and their cumulative grade point average (GPA). The results suggest that the academic library plays an important role in student learning and academic success. de Jager (1997) also reported that students with better academic performance tended to use more library materials than those with low grades. In a study analyzing the relationship between library use and academic performance, Allison (2015) found a correlation between greater use of an academic library and increase in GPA. The study also found that library use was correlated with student retention, providing evidence that the library played a positive role in student performance and adaptation to school. Soria, Fransen, and Nackerud (2014) found that first-year undergraduate students who used the library had higher GPA and higher retention than those who did not use the library. Specifically, library use areas, including database logins, book loans, and electronic journal logins, were positively associated with students' GPA and retention. The findings imply that library use increases students' access to academic resources and enhances their competence in academic work, thus leading to higher academic achievement.

In the previous literature, a range of studies reported an overall increase in academic library use, although some reported a decline in the use of certain library facilities and services. In addition, differences in academic library use by user characteristics have been reported. Some studies reported the positive relationship between library use and academic performance, demonstrating an important role of academic libraries in students' academic success.

METHOD

INSTRUMENT

A survey instrument was developed to measure user perception and patterns of use of an academic library. It included questions about the demographics of the respondents, their use of the university library and the library websites, and their perceptions of the library. The first section of the survey solicited information on the respondents' use of the university library, including questions about which places were preferred to study and read, which places were preferred to spend time during non-class times on campus, how frequently they visited the university library, and how long on average they stayed in the library at each visit to the library. In addition, the frequency of the activities that the respondents did in the library was measured using a 7-point Likert scale. To explore the relationship between physical library use and library website use, the second section of the survey included questions on the respondents' use of the university library websites. This included questions about whether they had ever visited the library websites, how frequently they visited the websites, and what services on the library websites they used most frequently.

Table 4
Use of the library websites.

Library website use		Frequency	Percentage
Experience with the library websites (n = 141)	Yes	119	84.4
	No	22	15.6
Frequency of library website visits (n = 119)	Almost everyday	16	13.4
	A few times a week	33	27.7
	A few times a month	37	31.1
	About once a month	16	13.4
	A few times a year	10	8.4
	About once a year	1	0.8
	Seldom	6	5.0

The third section included questions about the respondents' perceptions of the academic library. The study participants were asked to indicate what place they thought the university library was. The items were elicited based on Peterson's (2005) explanation of the types of user activity for which library space should be designed. In the present study, the study participants were asked to indicate how strongly they agreed with each of the items describing the university library as a place for information seeking, learning, meeting and socializing, recreation, or contemplation. The items were rated on a 7-point Likert rating scale (1 = strongly disagree, 2 = moderately disagree, 3 = slightly disagree, 4 = neutral, 5 = slightly agree, 6 = moderately agree, and 7 = strongly agree). The last section of the survey solicited information on the respondents' demographic information.

PROCEDURE

The study was conducted at a campus of a state university in the eastern region of the United States in February 2015. The campus offers 80 upper-level undergraduate and graduate degree and certificate programs on a full-time, part-time, day, evening, and weekend basis. It serves >4000 students including undergraduate and graduate students. The university library is located on the first floor of the building that was designed as the university's academic center in 2007. The library facility provides the equipment and study areas including group study rooms, a reading room, individual study carrels, and a study lounge.

This study was situated within wider academic research that was aimed at understanding user perception of an academic library. The data collection began after the approval of the research project by the university institutional review board (date of approval: January 27, 2015). Questionnaires were distributed personally to users in the lobby in front of the university library from 9:00 a.m. to 4:00 p.m. on 6th (Friday) and 12th (Thursday) of February 2015. The lobby, which was located on the first floor of the university's academic center, was the main entrance to the library. Study participants who had experience of using the university library were asked to fill out the questionnaire. A screening question was asked to confirm that they had experience of using the university library. The study participants who were at least 18 years of age or older were eligible for the survey. Participation was voluntary, and participants of the study were given a small gift card as an incentive for their participation. Of 160 questionnaires distributed, 147 questionnaires were received. After discarding incomplete responses, 141 responses were analyzed using descriptive statistics.

All statistical analyses were performed using SPSS version 22.0. Spearman's correlation analysis was performed to analyze the correlation of academic library use and user perception of the library.

RESULTS

Descriptive statistics were calculated for the characteristics of the respondents. The majority of the respondents were undergraduate students (65.2%) comprising freshmen (0.7%), sophomores (5.0%), juniors (30.5%), and seniors (29.1%). The remaining respondents included graduate students (24.1%), staff (5.7%), others (3.5%), and faculty (1.4%). In

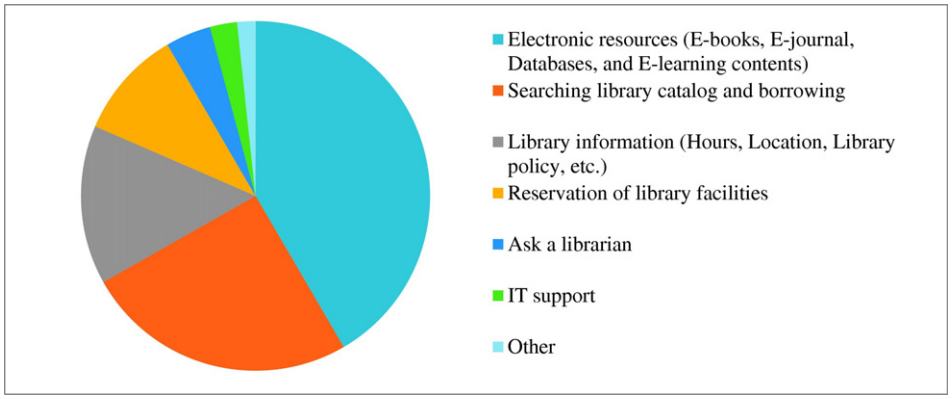


Fig. 1. The most frequently used services on the library websites (In descending order).

terms of gender, 67.4% of the respondents were female, and 32.6% were male. Regarding residence, the majority of the respondents resided off campus (97.2%), while a small percentage of the respondents resided on campus (2.8%).

PREFERRED PLACES

To identify which places were preferred by the respondents to study or spend time on campus, their preferences were examined. Regarding the preferred places to study and read, the study participants were asked to indicate three items and rank them according to their priority. For their first preferred place to study and read, a large number of the respondents indicated home (46.1%) and library (43.3%), followed by classroom (3.5%), lab (2.8%), and café (2.8%). The remaining respondents indicated dorm (0.7%) or lawn and outdoor bench (0.7%) as their first preferred place. For the second preferred place to study and read, a large number of the respondents indicated library (39.0%) and classroom (23.4%), followed by home (15.6%), café (10.6%), lab (6.4%), lawn and outdoor bench (4.3%), and other (0.7%). See Table 1 for additional details.

Regarding the preferred places to spend time during non-class times on campus, the study participants were asked to indicate three items and rank them according to their priority. The most preferred place to spend time during non-class times on campus was library (33.3%), followed by café (31.2%), classroom (11.3%), and gym (7.8%). The remaining respondents indicated lawn and outdoor bench (5.7%), lab (5.0%), dorm (2.8%), or other (2.8%) as their first preferred place. For the second preferred place to spend time during non-class times on campus, a large number of the respondents indicated library (26.2%), classroom (22.7%), and café (21.3%). The remaining respondents indicated lab (8.5%), gym (7.8%), lawn and outdoor bench (7.8%), sport turf (3.5%), other (1.4%), or dorm (0.7%). See Table 2 for additional details.

ACADEMIC LIBRARY USE

To explore the respondents' overall use patterns at the university library, the frequency of library visits and the average length of each visit were analyzed (see Table 3). Regarding the frequency of library visits, a large number of the respondents reported visiting the library a few times a week (35.5%), almost every day (29.8%), or a few times a month (17.7%). The remaining respondents reported visiting the library rather infrequently: about once a month (7.1%), seldom (6.4%), or a few times a year (3.5%). Regarding the question of how long on average they stayed in the library at each visit, a large number of the respondents reported staying for two to four hours (36.9%) or one to two hours (34.0%). The remaining respondents reported staying for less than an hour (12.8%), four to six hours (12.1%), or more than six hours (4.3%).

To identify the respondents' use of the library websites, the frequency of library website visits and the frequently used services on the websites were analyzed. The study participants were asked to indicate if they had ever visited the library websites. A large number of the respondents (84.4%) reported having visited the library websites, whereas the other 15.6% reported that they did not (See Table 4). Regarding the frequency of library website visits, a large number of the respondents indicated having visited the library website a few times a month (31.1%), a few times a week (27.7%), almost every day (13.4%), or approximately once a month (13.4%). The remaining respondents indicated 'a few times a year' (8.4%), 'seldom' (5.0%), or 'about once a year' (0.8%). Regarding the frequently used services on the library websites, the respondents were asked to choose two items among those provided. As presented in Fig. 1, the greatest percentage of the respondents indicated electronic resources (e-books, e-journals, databases, and e-learning contents) (41.6%), followed by searching the library catalog and borrowing (25.2%), library information (hours, locations, library policy, etc.) (14.7%), reservation of library facilities (10.1%), ask a librarian (4.2%), IT support (2.5%), and other (1.7%).

Spearman's correlation analysis was conducted to explore the correlation of use patterns of the physical library and the library websites. A correlation matrix is presented in Table 5. A significant positive correlation was found between the frequency of library visits and the average length of each visit to the library ($r = 0.38$, $p < 0.01$). It was also found that there was a significant positive correlation between the frequency of library visits and the frequency of library website visits ($r = 0.27$, $p < 0.01$). However, no significant correlation was found between the average length of each visit to the physical library and the frequency of library website visits ($p > 0.05$).

USER ACTIVITIES IN THE ACADEMIC LIBRARY

Regarding the activities users engage in while in the academic library, the study participants were asked to indicate the frequency of doing each of the activities using a 7-point Likert rating scale (1 = extremely infrequently, 2 = fairly infrequently, 3 = somewhat infrequently, 4 = neutral, 5 = somewhat frequently, 6 = fairly frequently, and 7 = extremely frequently). Table 6 presents the means and standard deviations of the frequency of user activities. The respondents

Table 5
Correlation coefficients of academic library use.

Variable	1	2	3
1. Frequency of library visits	1.00		
2. Average length of each visit to the library	0.38*	1.00	
3. Frequency of library websites visits	0.27*	0.18	1.00

* $p < 0.01$.

Table 6
Frequency of user activities in the academic library.

Activity	Mean	Standard deviation
Individual study	5.47	1.75
Reading	5.14	1.78
Computer use	5.10	1.93
Finding information	4.58	1.84
Group study	4.30	1.98
Using reference and information services	3.75	1.96
Meeting and socializing	3.69	1.91
Circulation	3.33	1.74
Eating and drinking	3.08	2.07
Taking class	3.01	2.33
Using audiovisual materials	2.78	1.95
Workshops or seminars	2.62	1.75
Participating in cultural events	2.22	1.55

indicated individual study (M = 5.47) as the most frequent activity, followed by reading (M = 5.14), computer use (M = 5.10), finding information (M = 4.58), group study (M = 4.30), using reference and information services (M = 3.75), and meeting and socializing (M = 3.69). User activities with relatively low means included circulation (M = 3.33), eating and drinking (M = 3.08), taking class (M = 3.01), using audiovisual materials (M = 2.78), workshops and seminars (M = 2.62), and participating in cultural events (M = 2.22).

Spearman's correlation analysis was performed to explore if the respondents' use patterns of the library were correlated with the frequency of each activity. The results from the correlation analysis are presented in Table 7. The frequency of library visits had a significant positive correlation with individual study ($r = 0.36$, $p < 0.01$), group study ($r = 0.29$, $p < 0.01$), reading ($r = 0.26$, $p < 0.01$), and computer use ($r = 0.17$, $p < 0.05$), implying that users who visit the library frequently are likely to engage in these activities more frequently. The average length of each visit to the library had a significant positive correlation with reading ($r = 0.40$, $p < 0.01$), individual study ($r = 0.39$, $p < 0.01$), eating and drinking ($r = 0.29$, $p < 0.01$), taking class ($r = 0.29$, $p < 0.01$), and computer use ($r = 0.19$, $p < 0.05$), implying that users who stay longer in the library at each visit are likely to engage in these activities more frequently. The frequency of library website visits had a significant positive correlation with using reference and information services ($r = 0.30$, $p < 0.01$), circulation ($r = 0.25$, $p < 0.01$), and reading ($r = 0.20$, $p < 0.05$), implying that users who visit the library websites frequently are more likely to engage in these activities.

USER PERCEPTION OF THE ACADEMIC LIBRARY

Regarding user perception of the academic library, the study participants were asked to indicate what place they thought the university library was. The results showed that the respondents perceived the academic library mainly as a place for learning (M = 6.27) and information seeking (M = 6.26), as indicated by their high mean scores (see Table 8). The mean scores for the library being a place for contemplation (M = 4.94) and a place for meeting and socializing (M = 3.70) were relatively high, whereas the mean score for a place for recreation was relatively low (M = 2.90).

To explore the relationship between the respondents' use patterns and their perceptions of the academic library, Spearman's correlation analysis was conducted. A significant positive correlation was found between the frequency of library visits and user perception of the library as a place for learning ($r = 0.21$, $p < 0.05$) (see Table 9). It implies that users who visit the library frequently are more likely to perceive the academic library as a place for learning. It was also found that there was a significant positive correlation between the average length of each visit and user perception of the library as a place for learning ($r = 0.29$, $p < 0.01$). It suggests that users who stay longer on average at each visit to the library are more likely to perceive the library as a place for learning.

DISCUSSION

The findings of this study suggest that the academic library is a preferred place to study and spend time during non-class times on campus. They support the findings of other studies that identified students' favorite places. Houlihan (2005) pointed out that the library became a more popular place to work and socialize, especially after allowing for food and drinks in the library. Waxman, Clemons, Banning, and McKelfresh (2007) found that the academic library was a popular place to meet fellow students, work on group projects, or find relaxation and restoration. In a study investigating students' use of the space in an academic library, Bryant, Matthews, and Walton (2009) reported that undergraduates tended to conduct academic work and social activities simultaneously. The open-plan space of the library that supported both academic and social activities was found to be a popular area.

In the present study, the frequency of library visits was found to be significantly and positively correlated with the average length of each visit to the library. The result implies that users who visit the library more frequently are likely to stay longer at each visit. In addition, it was found that the frequency of library visits was significantly and positively correlated with the frequency of library website visits. It suggests that users who visit the physical library frequently are more likely to visit the library websites frequently. However, there was no significant correlation between the average length of each visit to the physical library and the frequency of library website visits.

The most frequent user activity in the library was found to be individual study, followed in descending order by reading, computer use,

Table 7
Correlation coefficients between library use and user activities.

Variable	Frequency of library visits (n = 141)	Average length of each visit to the library (n = 141)	Frequency of library website visits (n = 119)
Individual study	0.36**	0.39**	−0.03
Reading	0.26**	0.40**	0.20*
Computer use	0.17*	0.19*	0.01
Finding information	0.05	0.16	0.16
Group study	0.29**	0.05	0.16
Using reference and information services	0.02	0.11	0.30**
Meeting and socializing	0.07	0.10	−0.11
Circulation	0.03	0.12	0.25**
Eating and drinking	0.11	0.29**	−0.04
Taking class	0.16	0.29**	0.05
Using audiovisual materials	−0.04	0.03	0.12
Workshops or seminars	−0.07	0.05	0.12
Participating in cultural events	−0.02	0.07	0.12

* $p < 0.05$.
** $p < 0.01$.

Table 8
User perceptions of the academic library.

Perception of the academic library	Mean	Standard deviation
Place for learning	6.27	1.10
Place for information seeking	6.26	1.08
Place for contemplation	4.94	1.62
Place for meeting and socializing	3.70	1.86
Place for recreation	2.90	1.75

finding information, group study, using reference and information service, and meeting and socializing. The results are consistent with other studies reporting that study space is the most frequently used space in academic libraries. King, Buss, Cohen, Stanley, and White (2008) emphasized the importance of study space in that the most commonly used spaces in the library were spaces for individual and group study. Reporting on the trend of student use of library, Lawson (2004) noted that many used the academic library as a place to study individually or in groups. Thus, academic libraries need to provide a variety of study spaces that satisfy users' learning styles and space preferences. Reading is also a primary purpose for coming to the academic library. Freeman (2005) noted that many students considered the traditional reading room as their favorite area of the library. The use of public computers is still strong in libraries, and academic libraries are increasing the number of computers and peripherals (Lux, Snyder, & Boff, 2016; Whitmire, 2001). Finding information was also reported as a frequent activity conducted in the academic library. The mean scores for user activities such as using reference and information services and meeting and socializing were also found to be relatively high, thus implying the significance of the provision of effective library services and the welcoming environment that enriches users' social experiences.

Regarding the relationship between use patterns and user activities in the library, the frequency of library visits had a significant positive correlation with individual study, group study, reading, and computer use. It implies that users who visit the library frequently are likely to engage in activities such as individual study, group study, reading, and computer use more frequently. It suggests that the academic library needs to expand its facilities and services in support of these activities to attract more users to the library. The average length of each visit to the library had a significant positive correlation with reading, individual study, eating and drinking, taking class, and computer use. It implies that users who stay longer in the library at each visit are likely to engage in these activities more frequently. Given that users who engage in such activities tend to stay longer in the library, the library needs to create a comfortable and efficient environment to support the activities. In addition, the frequency of library website visits had a significant positive correlation with using reference and information services, circulation, and reading. It suggests that users who visit the library websites frequently are likely to engage in activities such as using reference and information services, circulation, and reading more frequently. Given the nonsignificant correlation between the frequency of library website visits and the average length of each visit to the library, the frequent website visitors who do not stay for a long time in the library need to be provided with efficient and streamlined services when they visit the library.

Regarding user perception of the academic library, the findings of this study suggest that users perceive the academic library primarily as a place for learning and information seeking. As learning in higher education emphasizes group interaction and discussion that enable students to discover and share knowledge with other learners (Sens, 2009), academic libraries need to support the social and collaborative aspects of students' learning processes by reshaping learning spaces. Accommodating the new approach to learning and a variety of users' learning styles, academic libraries should make efforts to reshape library space accordingly.

The mere provision of study space may not facilitate users' learning processes. Along with the development of space and facilities, the provision of information and services is essential in academic libraries to enhance learning. It may include the provision of print and digital collections that satisfy information quality requirements and user preferences, the provision of course reserves, and the effective management of the output from the knowledge activities of academic communities. Academic libraries can play an active role in facilitating the activities of learning communities and communities of practice (CoPs) in their academic community and organizing the output from their activities. To serve as a place for information seeking, academic libraries need to provide effective information services and programs that facilitate the learning and knowledge activities of the academic community. In addition, the mean scores for user perceptions of the academic library as a place for contemplation and a place for meeting and socializing were also relatively high, implying that the library is perceived as a multi-purpose destination that is expected to serve multiple functions embracing various user activities.

CONCLUSION

This study provides the empirical evidence of user perception and use of the academic library. It suggests that the library is a preferred place to study and spend time on campus. It was found that the frequent user activities in the library included individual study, reading, computer use, finding information, group study, using reference and information service, and meeting and socializing. Users tended to perceive the academic library primarily as a place for learning and information seeking. User perception of the academic library as a place for contemplation and a place for meeting and socializing was also found to be relatively high, implying users' expectation of the library as a multi-purpose destination that supports various user activities. Given the importance of learning and information seeking that is identified in user perception of the library, academic libraries need to accommodate user space for learning and provide effective information services and programs. To remain a valued place for the academic community, libraries should evolve into an attractive place to support a range of user activities by continuously reforming their facilities and services.

The findings of this study show how users in current higher education perceive the academic library and how they use the library space and services. They may provide insights into how academic libraries can strengthen their place and advance their role in the academic

Table 9
Correlation coefficients between library use and perception of the academic library.

Variable	Frequency of library visits (N = 141)	Average length of each visit to the library (N = 141)	Frequency of library website visits (N = 119)
Place for learning	0.21*	0.29**	0.12
Place for information seeking	−0.07	−0.01	0.10
Place for contemplation	−0.10	−0.06	0.08
Place for meeting and socializing	−0.06	0.05	−0.06
Place for recreation	−0.07	−0.09	−0.00

* $p < 0.05$.
** $p < 0.01$.

community. There are some limitations of the study. This study was conducted at a university where most students resided off campus. Future research can be conducted in other universities to obtain more generalizable results. In addition, further studies are needed to examine user preferences for specific features of library space designed to support various user activities.

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